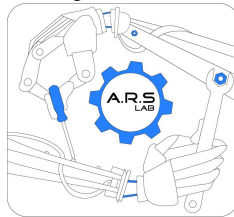


Logic Circuits and Signals

Hardware/Software Connection

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Embedded Systems Course

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Logic Circuits and Signals

Notes

Logic Circuits

Notes

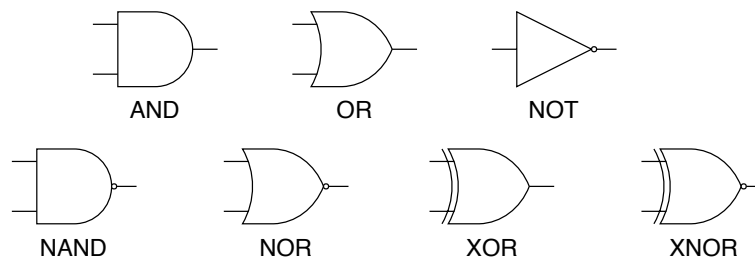
- Logic circuits are characterized by the fact that **voltage of wires** can assume only **two values**:
 - 0 V , corresponding to **Logic/bit-value “0”**
 - $+V_{DD}$, corresponding to **Logic/bit-value “1”**
- where V_{DD} is the power supply of the whole circuit, it may be 5 V , 3.3 V , 1.8 V

Notes

Logic Circuits Classification

Combinatorial Circuits

- They are logic circuits where the output is dependent only on the **current state** of the inputs
- **Logic gates** (AND, OR, NOT, etc.) are kind of combinatorial circuits
- Their behaviour is represented by a **truth table**

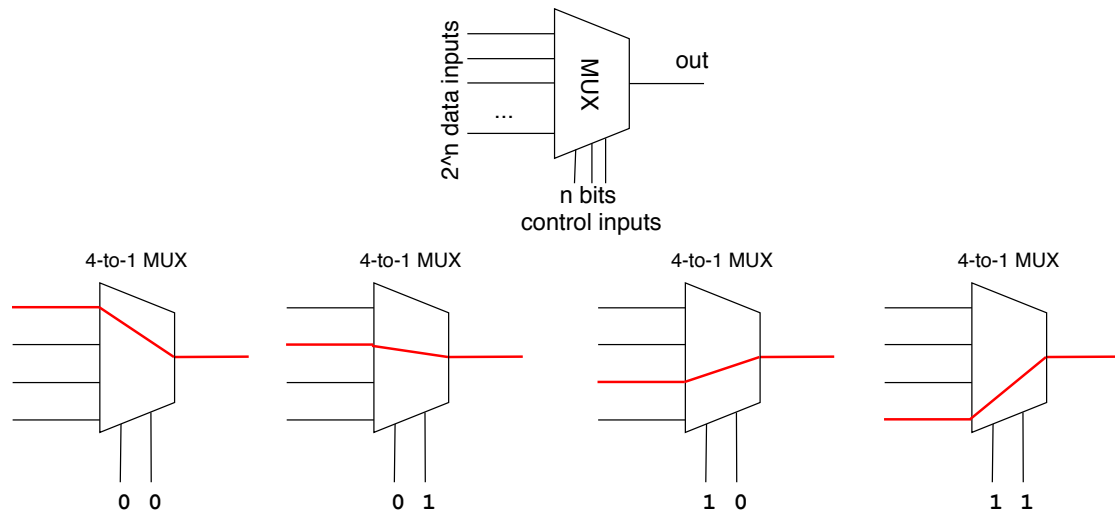


Notes

Special Combinatorial Circuits

Multiplexers

- **Multiplexers** are combinatorial circuits that act as **signal switches**
- One **data input** at time is routed to the output on the basis of the values of **control input bits**



Notes

Sequential Circuits

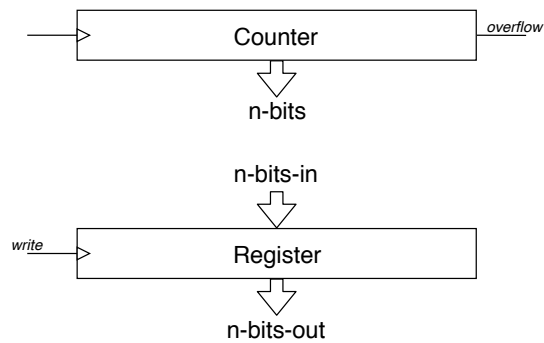
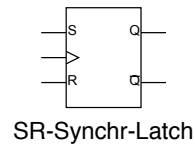
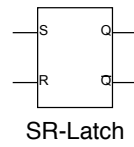
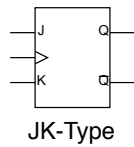
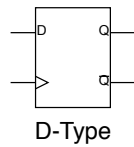
- They are logic circuits where the output is dependent only on the **current and past state** of the inputs
- They are sensible to **variations** of the inputs
- They are logic circuits that have a **memory**
- **Flip-Flops** and their derivatives are kind of combinatorial circuits
- Their behaviour is represented by a **finite-state machine**
- In general, a **clock signal** drives their behaviour

Notes

Logic Circuits Classification

Kind of Sequential Circuits

- Flip-flops
- Counters
- Registers



Notes

Logic Signals

Notes

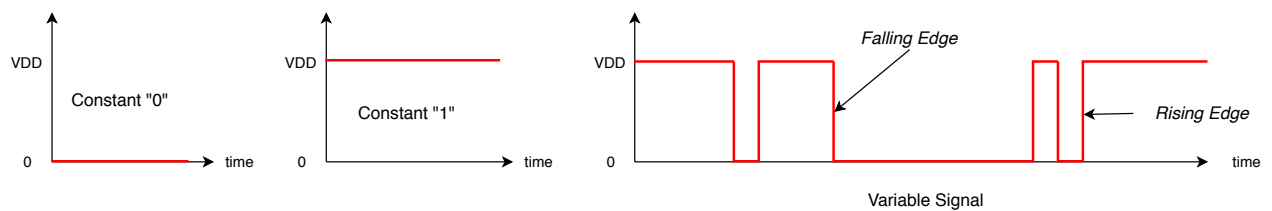
Logic Signals

Constant Signals

- They are logic states that does not vary in time

Variable Signals

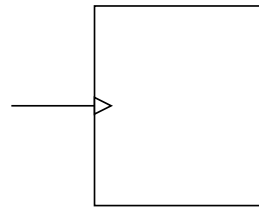
- They are logic states that may vary in time
- They are featured by **edges**
 - **Falling Edge**: variation from “1” to “0”
 - **Rising Edge**: variation from “0” to “1”



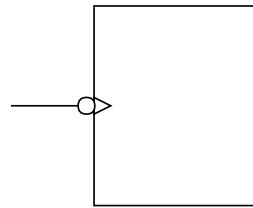
Notes

Edges and Sequential Circuits

- Sequential Circuits are sensible to **edges**
- Edge inputs are represented in circuits by “triangles”
- **simple triangle: rising edge**
- **circle + triangle: falling edge**



State change on
rising edge

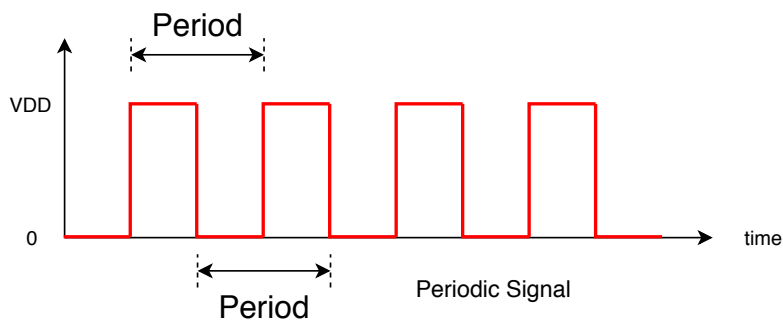


State change on
falling edge

Notes

Periodic Signals

- They are kind of *variable signals* where the **time distance** between **two edges of the same time** is **constant**
- This distance is called **Period, P** and measured in seconds
- The **frequency**, computed as $f = \frac{1}{P}$, is the **number of “periods” per second** and is measured in **Hertz, Hz**

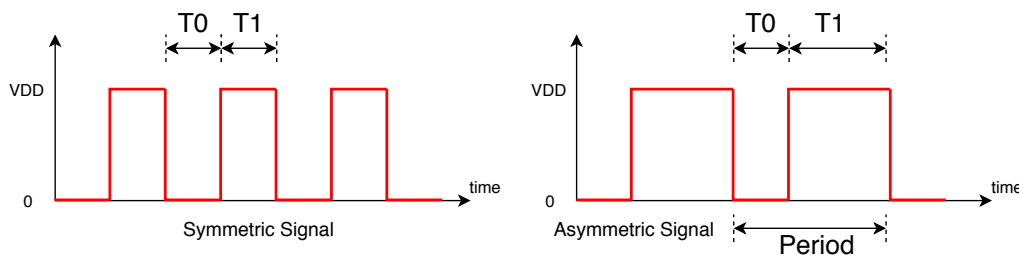


Notes

Periodic Signals

- Periodic signals can be:
 - **symmetric**: the time durations of state “0” and state “1” is the same and equal to $T_0 = T_1 = \frac{P}{2}$
 - **asymmetric**: the time durations of state “0” and state “1” is different $T_0 \neq T_1$
- The “asymmetry” is called **duty cycle** and is the **percentage of period in which the signal is “1”**

$$DC = \frac{T_1}{T_0 + T_1} 100 = \frac{T_1}{P} 100$$



Notes

Time and Frequency Measures

Time Measures

- **milliseconds:** $1\text{ ms} = 10^{-3}\text{ s}$
- **microseconds:** $1\text{ }\mu\text{s} = 10^{-6}\text{ s}$
- **nanoseconds:** $1\text{ ns} = 10^{-9}\text{ s}$
- **picoseconds:** $1\text{ ps} = 10^{-12}\text{ s}$

Frequency Measures

- **KiloHertz:** $1\text{ KHz} = 10^3\text{ Hz}$
- **MegaHertz:** $1\text{ MHz} = 10^6\text{ Hz}$
- **GigaHertz:** $1\text{ GHz} = 10^9\text{ Hz}$
- **TeraHertz:** $1\text{ THz} = 10^{12}\text{ Hz}$

Notes

Hardware/Software Connection

Notes

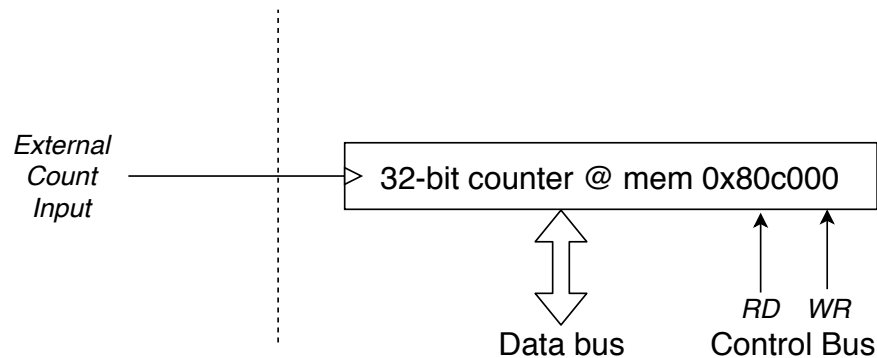
Hardware/Software Connection

- Circuits of a MCU are “connected” to the software through **registers**
- A **register** is mapped in memory at a **known memory address**
- Hardware/software interaction is performed by **reading/writing** at that **memory address**

Notes

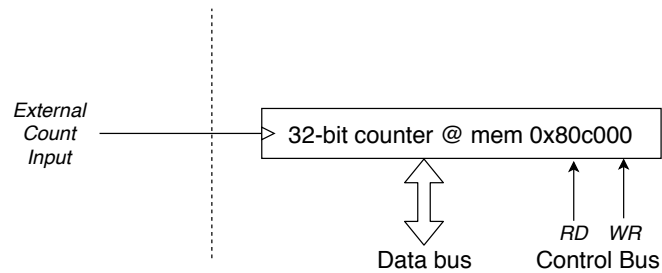
An Example: Piece Counter

- A “presence sensor” generates a pulse each time a “piece” is identified
- The pulse is connected to an **external counter** MCU input
- Each time the sensor generates a pulse, the counter increments (in hardware)



Notes

Hardware/Software Connection

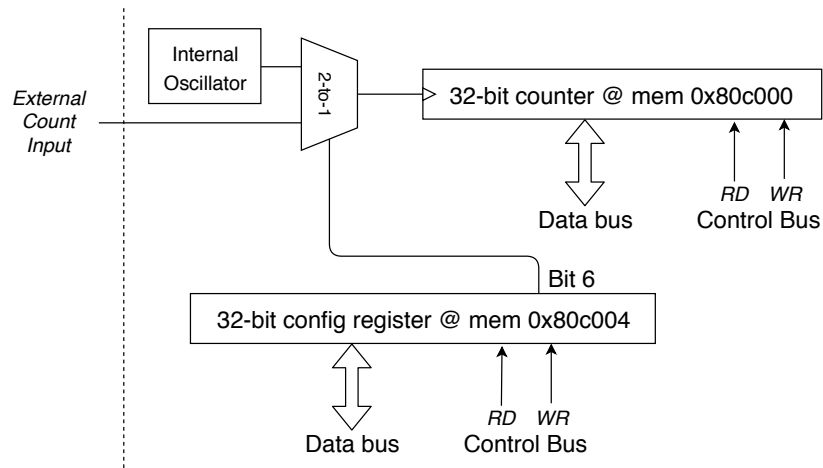


Reading/Writing the Counter

```
...  
int32_t * counter_ptr = (int32_t *)0x80c000;  
...  
  
/* clearing the counter */  
*counter_ptr = 0;  
...  
  
/* printing the counter */  
printf("Counter_value_%d\n", *counter_ptr);
```

Notes

Hardware/Software Connection



Configuring the Counter

```
int32_t * config_ptr = (int32_t *)0x80c004;

/* counter input from Internal Oscillator */
*config_ptr = (*config_ptr) & ~(1 << 6);

/* counter input from External Counter Input */
*config_ptr = (*config_ptr) | (1 << 6);
```

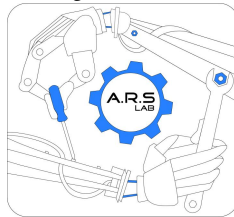
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